

User Research Project

The Jule Collins Smith Collection Database

Jule Collins Smith Museum of Fine Art
Curatorial & Communications Units

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Date: 11/7/2024

Subject: **Recommendation Report**

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Summary

In a usability study focused on evaluating The Jule Collins Smith Museum's (The Jule) front-facing collections page, a mixed-method research design was employed, combining structured survey questions with the Think-Aloud Protocol (TAP) method. The survey questions gathered quantitative data on participants' initial impressions, ease of navigation, content clarity, and overall satisfaction. Meanwhile, TAP allowed participants to vocalise their thoughts as they interacted with the site, providing deeper qualitative insights into their immediate reactions, challenges, and cognitive processes while navigating the website.

The results highlighted both strengths and areas for improvement in the website's usability. Key findings included user challenges in hierarchy and locating specific information on the website pages, as well as some design flaws which led to minor frustration and hesitancy in proceeding through the site. Overall, the integration of survey feedback with real-time participant observations through TAP provided a comprehensive understanding of user experiences, guiding actionable recommendations for enhancing the website's functionality and user satisfaction.

Recommendations to the website would include enhancing the searchability from the landing page to allow users to navigate more effectively and efficiently to get the information they require. Additionally, the landing page design could benefit from enhanced graphic communication, such as increased contrast and refined spacing, to improve visual clarity and guide user attention more effectively.

Introduction

The team was tasked with finding a client, and conducting a usability evaluation in a technological mediated space for this project. We decided that The Jule Museum would be a good client and submitted a project plan proposal which highlighted our client, their audience and the rationale for conducting research with the museum. Our focus led to conducting a usability research test on the museum's front-facing collections page. George & Estes Brewer, notes that, "research should involve the action of collection and analysis of data, create new knowledge, and should inform actions and decisions" (2020,pg.2). The purpose of this research project was to identify specific problem areas that users may have while using the client's newly-launched website, and gather insights to guide targeted improvements and recommendations for the client.

Within this report we highlight our users, their background demographics, and the methods which we employed in acquiring information from the research testing. Additionally, this report will further delve into the research findings based on the user tests conducted, highlight specific tasks which the participants had to complete and illustrate our findings through visual qualitative and quantitative data charts.

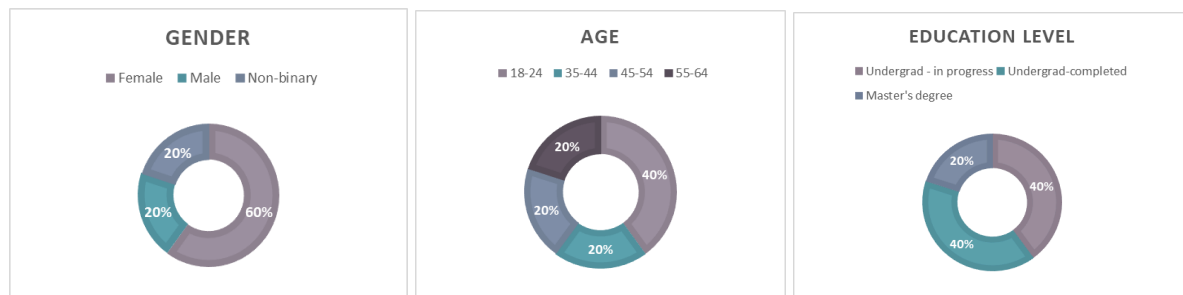
Based on the research results achieved, we highlight some key points in which users gave feedback on their experience while conducting this usability test on The Jule's website.

Methods

The primary objective of this usability study was to evaluate the user experience of The Jule Collins Smith Museum's front-facing collection page, identifying specific problem areas and gathering insights to guide targeted improvements and recommendations for the client.

Participant Profile:

Five participants were recruited for the study, representing a diverse range of ages, backgrounds, and levels of familiarity with the museum's website. The participants' demographics are as follows (Figure 1):



(Figure 1)

Age: 18–24 (2), 35–44 (1), 45–54 (1), 55–64 (1)

Background: Students and staff members, all individuals with varying levels of familiarity with the website.

Evaluation Methods: A mixed-method research design was employed, combining structured survey questions with the Think-Aloud Protocol (TAP) method. The survey questions gathered quantitative data on participants' initial impressions, ease of navigation, content clarity, and overall satisfaction. The TAP method allowed participants to vocalise their thoughts as they interacted with the site, providing deeper qualitative insights into their immediate reactions, pain points, and cognitive processes while navigating the website.

Evaluation Environment:

The usability tests were conducted in a controlled environment, at the museum's Study Room, with participants using their own electronic devices to access the museum's website.

Data Collection Methods and Tools:

The following methods and tools were used to collect data:

- Structured survey questions: Participants completed a questionnaire with open-ended fields and System Usability Scale (SUS) prompts.
- Think-Aloud Protocol (TAP): Participants were asked to vocalise their thoughts as they interacted with the website.
- Speech recording: Participants' vocalised interaction with the website was recorded using voice recording software.
- Note-taking: Researchers took notes during the usability tests to capture the participants' comments and observations.

Roles of Team Members:

Our three-membered team had the following roles assigned to them prior to testing sessions:

Jolisa: Took notes during tests observations for qualitative evaluation.

Jessica: Led the tests by reading the requisites and tasks to the participants with a script.

Muhammad: Recorded the think-aloud sessions for later transcript.

Results

In this section, we outline the findings from our user testing. We offer a breakdown of the task scenarios, including addressing specific sticking points, and suggest causations by task. We detail further the broader, overall user experience, including participant impressions and suggestions in their own words.

Task Results

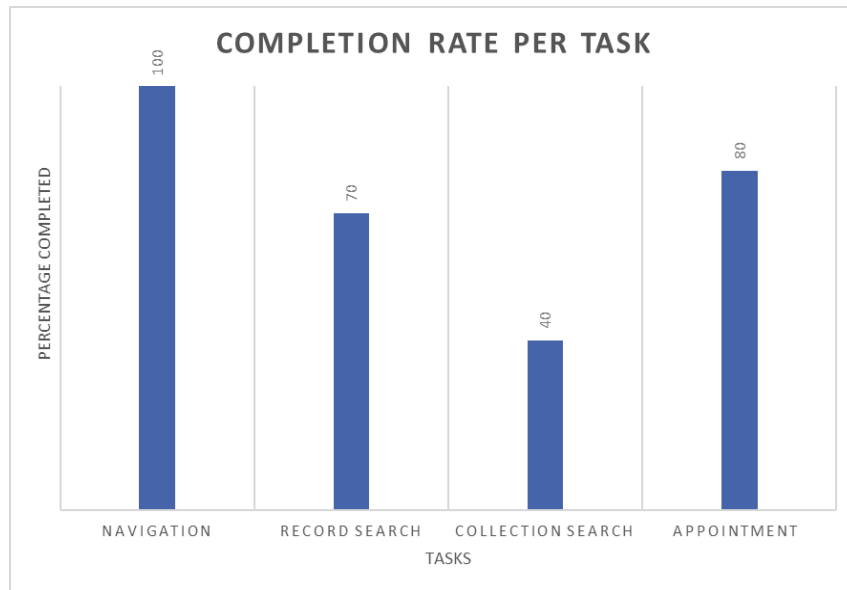
	participant 1	participant 2	participant 3	participant 4	participant 5	Average
Overall Task Completion Rate	87.5%	62.5%	50%	75%	87.5%	72.5%

(Figure 2)

Our findings indicate a 72.5% overall task completion rate (Figure 2) based on the following prompts:

- **Task 1:** Open jcsn.auburn.edu. From this landing page, find and navigate to the “Collection Database” page.
- **Task 2:** From the “Collection Database” page, find the record for John James Audubon’s Yellow-throated Vireo. What year was the print created?
- **Task 3:** Now, locate an object from the museum’s “Advancing American Art” collection. Which object have you selected and why?
- **Task 4:** How would you request an in-person appointment to view an object from the collection? (Don’t actually submit!)

Broken down by task, participant experience varied from 40–100% completion rate (Figure 3).



(Figure 3)

Task 1:

“Open jcsn.auburn.edu. From this landing page, find and navigate to the “Collection Database” page.”

For task 1, users had a 100% completion rate. All but one participant was able to quickly navigate to the Collection Database page. The participant who took the longest to find the appropriate page, was eventually able to complete the task, but with difficulty. This user, the oldest (age 55–64) of our test group, disliked having to open the hamburger menu for navigation options, expecting instead to find navigation menus across the top of the page.

This participant even missed the Collection Database link in the hamburger menu and tried several other options before finally locating the secondary link at the bottom of the homepage. Though the remaining users experienced far less difficulty with this task, more than one user commented that the link’s location was confusing or non-intuitive. These user experiences may indicate the need for additional graphic signalling, perhaps considering additional contrast and proximity design options. The repetition of the database link in the hamburger menu and at the bottom of the homepage is a useful duplication, with one in five of our participants navigating via the secondary (bottom of the homepage) link.

Task 2:

“From the “Collection Database” page, find the record for John James Audubon’s *Yellow-throated Vireo*. What year was the print created?”

For task 2, participants had a 70% completion rate. All five of the participants were able to complete some portion of the task, but three of the participants were unable to locate an object date. This task was designed to explore several aspects of the object record search process, and it exposed two primary usability considerations.

The first consideration involves user search preferences. Participants were directed with this task to find a specific object record, John James Audubon's *Yellow-throated Vireo*. Participants varied in their search paths. Three participants utilised the "title" search, one the "artist" search, and one participant used a combination of both the "title" and "artist" search fields (Figure 4).

Search flexibility is a key attribute of the database; however, participants did express some confusion with the search options. Two users expected to be able to search primarily by artist, not expecting the title search to feature as prominently on the page (despite eventually choosing to use the title search), and a third chose to search by artist from the onset. The usability of these search features was also addressed with more than one user highlighting functionality. One participant described the interface as "not very usable" and another found that the artist search option was "odd" because of its sorting by artist first, rather than last, name (Figure 5).

The image shows a search interface with several filter sections. The 'Title' section has a search box labeled 'Search by Title'. Below it are dropdown menus for 'Artist' (set to 'All Items'), 'Artist Nationality' (set to 'All Items'), 'Medium' (set to 'All Items'), and 'Credit Line' (set to 'All Items'). At the bottom is a green 'SUBMIT' button and a 'Reset' link.

(Figure 4)

The image shows a search results dropdown menu. It starts with 'All Items' at the top, followed by a search box. Below the search box is a list of items with counts: 'All Items', '"Sam G." Granger (1)', '(Chinese) Unidentified (4)', '(Indian) Unidentified (1)', '(Sino-Tibetan) Unidentified (15)', '(Staffordshire or Yorkshire Pottery) Unidentified (1)', '(Tibetan) Unidentified (22)', 'A. Breed (1)', and 'Abraham Munting (2)'. The list is scrollable.

(Figure 5)

A second consideration focuses on search results. Multiple participants noted that more than one record for Audubon's *Yellow-throated Vireo* populated in their search (Figure 6). Participants were not able to easily differentiate between these records on the search page due to similarities in the content provided and the lack of additional distinguishing details (such as artist name). Upon selecting an object record, three of the five participants were unable to locate the object's date. Content inconsistencies (some date fields appear to not be populated) and the small size of the date text relative to the other content (Figure 7) appears to have contributed to these user issues.

Collection Database

Found 3 Results
Page 1 of 1

Title
yellow throated

Artist
All Items

Artist Nationality
All Items

Medium
All Items

Credit Line
All Items

SUBMIT

Reset



Yellow-throated Vireo



Yellow-throated Vireo



Yellow-throated Vireo,
Blue-winged Warbler,
Great Horned Owl

Page 1 of 1

(Figure 6)



(Figure 7)

Task 3:

“Now, locate an object from the museum’s “Advancing American Art” collection. Which object have you selected and why?”

The completion rate for task 3 was 40%, the lowest of all the task scenarios. Those who were successful, two of our five participants, in locating an object from a specific collection were familiar enough with museum practices to know to search by credit line. The remaining participants searched using the title field and were unsuccessful, as the title search is not a general search field. (One successful participant even anticipated these search problems, suggesting that for those less familiar with the museum, the task might be more difficult because they might assume they should search by title.)

All participants were able to work through resetting the search (either through the “reset” button or by selecting the back button on their browser) from the previous task or from failed in-task searches by collection, but with some difficulty. Participants

found themselves restarting searches or reloading results multiple times while working through the search process.

Task 4:

“How would you request an in-person appointment to view an object from the collection? (Don’t actually submit!)”

For task 4, the completion rate was 80%. Four of the five participants were able to find the appropriate form to schedule an art viewing under the “Plan a Visit” menu. Participants did experience slight hesitation when looking through the hamburger menu, with one participant incorrectly beginning their search under the “Education” menu before locating the correct menu option. The one unsuccessful participant (from our age 55–64 demographic) assumed the homepage would include a general inquiry phone number or email address for scheduling. While they were able to find a staff directory under the “About” section (“in small print”), they were not sure how to proceed with this task or even with a general inquiry, concluding that they would plan to email a staff member at random for assistance.

Overall User Experience

At the conclusion of the above task scenarios, our team also asked participants to complete a questionnaire with open-ended fields and SUS prompts. The following results are in the participant’s own words.

Liked Most	Liked Least
“the sleek look of it and use of artwork images”	"the database does not allow for a general search"
“the new, more focused search tools in the collection database”	"navigation"
"colourful"	"it was fairly difficult to find where to request a tour or to see an object"
"easy to navigate"	"images don’t load correctly sometimes, auto search while typing"
"the collection database is easier to find"	

Impressions

"I have used the website since the redesign. I like the new look of it. It feels more modern and more like how I would anticipate a university art museum to look."

"Not intuitive or easy to find relevant information"

"Some of the user interface seems difficult to use. Not all of the site features are in the place I would assume them to be. I think the collections database should have a more general search, and not be so niche or detailed"

"In order to request to see an object I had to scroll quite a ways down the page. It was not particularly intuitive."

Suggestions for Improvement

"Have a general search bar for collections database"

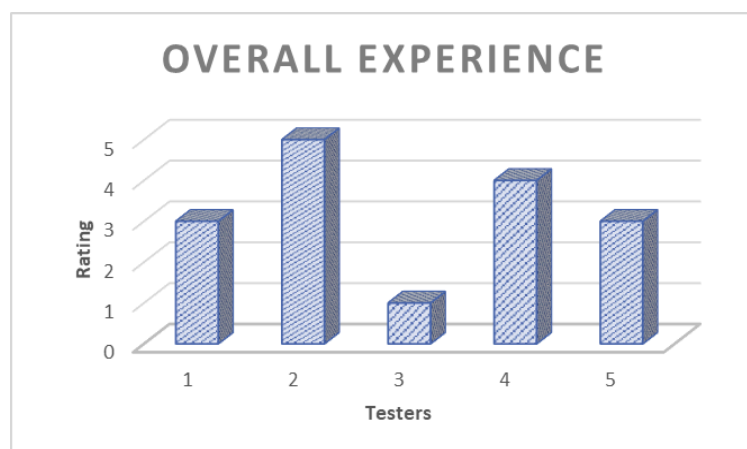
"Have more easily recognizable places to find the features"

"Remove the auto search when typing"

"More information across the top of the page"

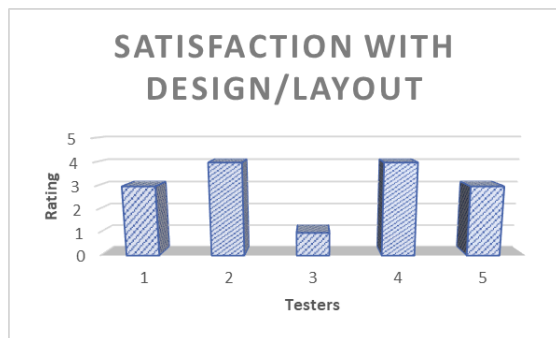
System Usability Scale Results

Average overall experience rating of 3.2 out of 5 (Figure 8).

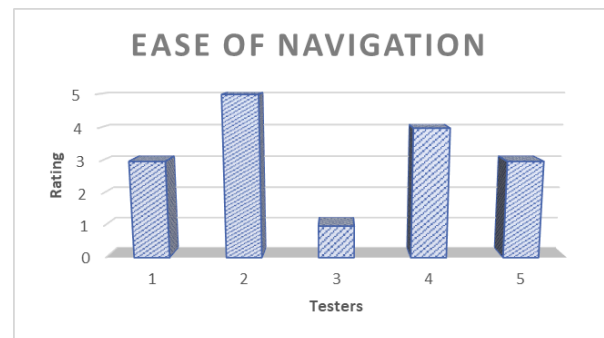


(Figure 8)

Average satisfaction with design/layout rating of 3 out of 5 (Figure 9).

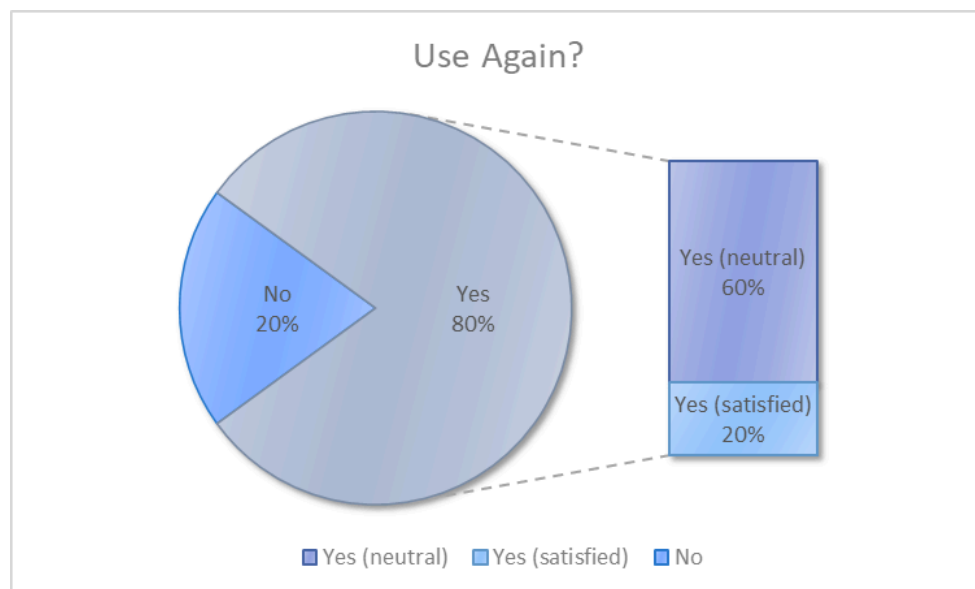


Average ease of navigation rating of 3.2 out of 5 (Figure 10).



(Figures 9 & 10)

4 out of 5 testers said that they would use the database again (Figure 11).



(Figure 11)

Recommendations

Based on the findings detailed above, our team has identified some key areas for improvement to enhance the overall user experience. The following recommendations are designed to address the challenges encountered by participants during the study.

Search Functionality

1. **Introduce keyword search:** System developers should consider providing a keyword search option beyond the existing categories to enable users to search the target item or category with more flexibility.

2. **Include search instructions or tips:** Incorporate guidance on how to use the search function effectively, including any specific search operators or syntax.

Collection Database

1. **Enhance search results:** Improve the display of search results to facilitate easier differentiation between similar records, including the addition of distinguishing details such as artist names.
2. **Increase font text size:** Enlarge some of the font text which provides important information to increase easier readability and reduce user frustration.
3. **Address content inconsistencies:** To address the issue of multiple indistinguishable records populating for users during specific searches, it is recommended to add distinguishing details to each record on the search results page. This could include information like the artist's name, object date, or other unique identifiers. Implementing these additional details would help users quickly differentiate between records and select the one they need without confusion.

Accessibility and Usability

1. **Ensure accessibility:** Administrators of the website must ensure that the accessibility of the website meets the comprehensive standard benchmarks implemented on any facility's website.
2. **Conduct user testing with diverse audiences:** System admins must conduct regular usability tests with diverse user groups, including older adults and individuals with varying levels of accessibility needs. This helps to identify and address potential issues, reduce the number of "clicks" it takes to get through a simple task, and navigate the website page more efficiently to find information such as finding artist's name or a particular artifact.

References

George, H. F., & Estes Brewer, P. (2020). *A Research Primer for Technical Communication: Methods, Exemplars and Analyses*.